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AGILE DEVELOPMENT AND METHODOLOGIES

LAB FAT

13.04.26

Q: EXPENSE TRACKER SYSTEM

App.java

src > main > java > App.java

```java
public class App {
    private String category;
    private double amount;

    public App(String category, double amount) {
        this.category = category;
        this.amount = amount;
    }

    public String getCategory() {
        return category;
    }

    public double getAmount() {
        return amount;
    }
}
```

AppTest.java

src > test > java > AppTest.java

```java
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class AppTest {

    @Test
    void testConstructor() {
        App expense = new App("Food", 200);

        assertEquals("Food", expense.getCategory());
        assertEquals(200, expense.getAmount());
    }

    @Test
    void testGetCategory() {
        App expense = new App("Travel", 500);

        assertEquals("Travel", expense.getCategory());
    }

    @Test
    void testGetAmount() {
        App expense = new App("Shopping", 1000);

        assertEquals(1000, expense.getAmount());
    }

    @Test
    void testDifferentValues() {
        App expense = new App("Bills", 300);

        assertNotEquals("Food", expense.getCategory());
        assertNotEquals(100, expense.getAmount());
    }
}
```

Running mvn clean package:

```
[INFO] Loaded 22537 auto-discovered prefixes for remote repository central (prefixes-central.txt)
[INFO] Loaded 74 auto-discovered prefixes for remote repository apache.snapshots (prefixes-apache.snapshots-c1d4c55f9308e5ac18a4
1dca64d85c515.txt)
[WARNING] Version not locked for default bindings plugins [maven-clean-plugin, maven-resources-plugin, maven-jar-plugin], you sh
fine versions in pluginManagement section of your pom.xml or parent
[INFO]
[INFO] --- clean:3.4.0:clean (default-clean) @ simple-java-app ---
[INFO] Deleting C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app\src\main\resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ simple-java-app ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app\target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ simple-java-app ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app\target\test-classes
[INFO]
[INFO] --- surefire:3.0.0-M7:test (default-test) @ simple-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.164 s - in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ simple-java-app ---
```

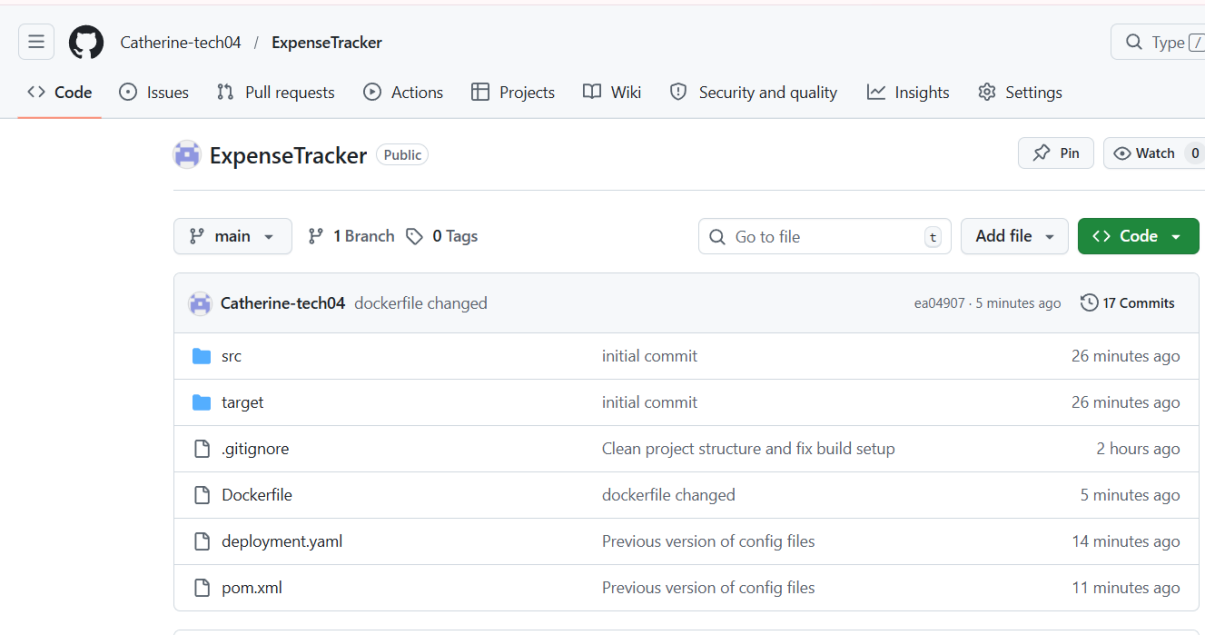
Running mvn test package:

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.302 s - in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app\src\main\resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ simple-java-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-java-app ---
[INFO] skip non existing resourceDirectory C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ simple-java-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:3.0.0-M7:test (default-test) @ simple-java-app ---
[INFO] Skipping execution of surefire because it has already been run for this configuration
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ simple-java-app ---
[INFO] Copying com.demo:simple-java-app:pom:1.0 to project local repository
[INFO] Copying com.demo:simple-java-app:jar:1.0 to project local repository
[INFO] Copying com.demo:simple-java-app:pom:consumer:1.0 to project local repository
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.824 s
[INFO] Finished at: 2026-04-13T10:53:49+05:30
[INFO] ------------------------------------------------------------------------
```

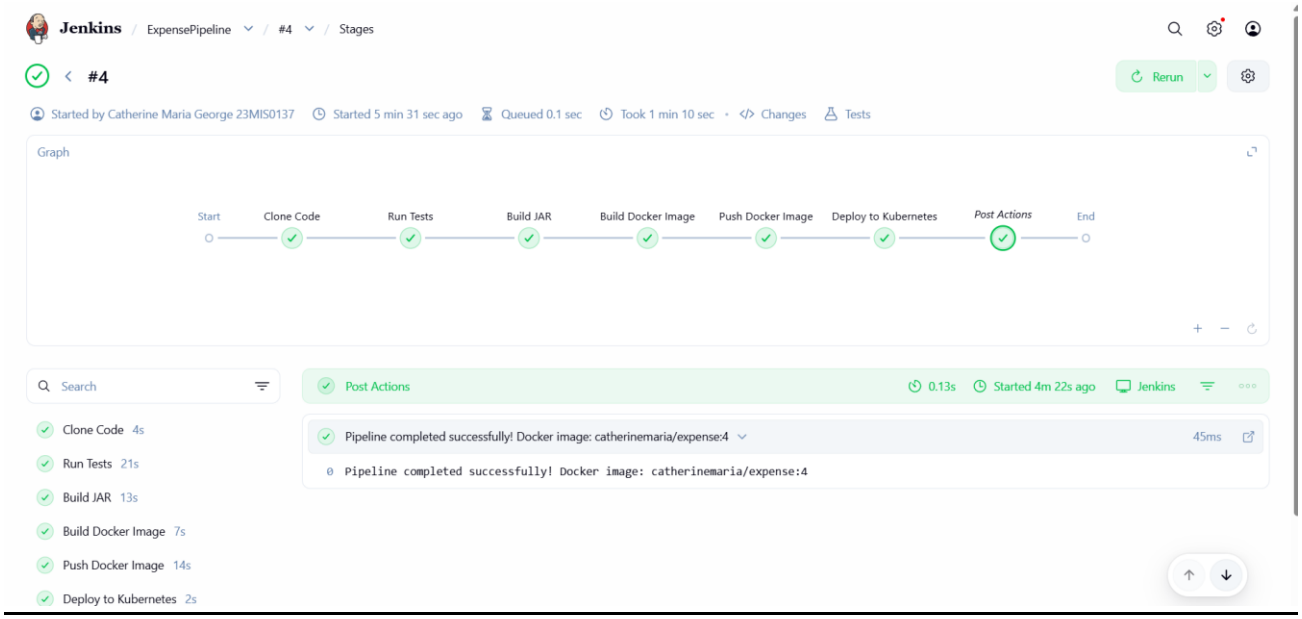
Git directory structure:

```
PS C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app> dir


    Directory: C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        02-04-2026     13:44                src
d-----        13-04-2026     10:10                target
-a----        13-04-2026     08:35             20 .gitignore
-a----        13-04-2026     10:21            546 deployment.yaml
-a----        13-04-2026     10:32            125 Dockerfile
-a----        13-04-2026     10:27           1303 pom.xml


PS C:\Users\cathu\OneDrive\Desktop\Sem 6\Agile\simple-java-app>
```

GitHub:

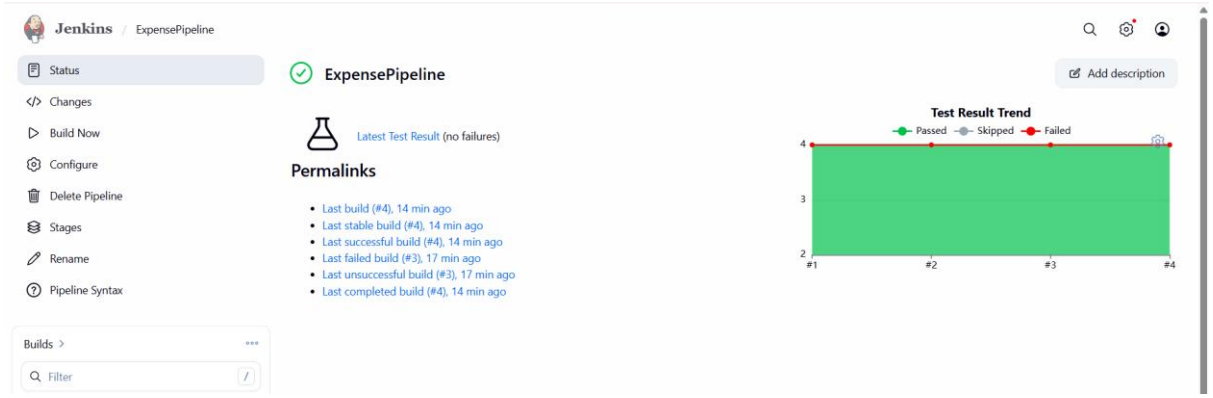
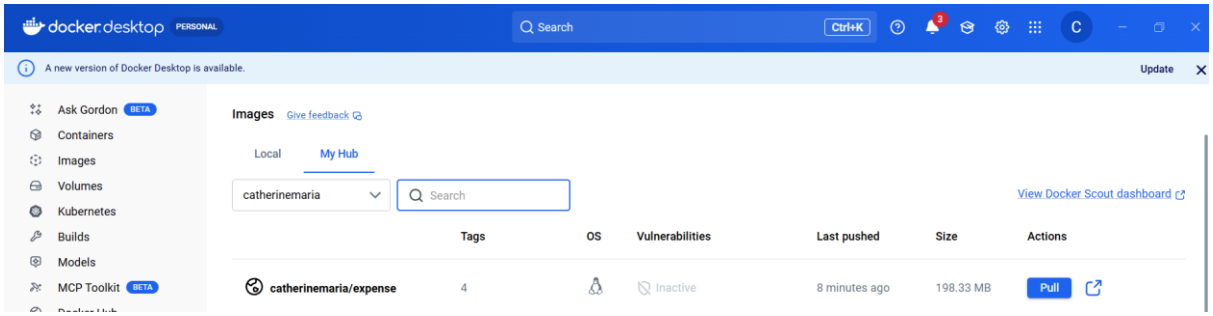


Pipeline on Jenkins:



Console output:

```
C:\ProgramData\Jenkins\.jenkins\workspace\ExpensePipeline>kubectl config current-context
docker-desktop

C:\ProgramData\Jenkins\.jenkins\workspace\ExpensePipeline>kubectl get nodes
NAME             STATUS   ROLES           AGE   VERSION
docker-desktop   Ready    control-plane   10h   v1.34.1

C:\ProgramData\Jenkins\.jenkins\workspace\ExpensePipeline>REM Apply deployment YAML

C:\ProgramData\Jenkins\.jenkins\workspace\ExpensePipeline>kubectl apply -f deployment.yaml --validate=false
deployment.apps/java-app configured
service/java-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed successfully! Docker image: catherinemaria/expense:4
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



On DockerHub: